Diploma in Electronics / Electronics & Communication Engineering • Semester 3

Electric Circuits & Networks

Full-width desktop lesson sample with textbook-style explanation, diagrams, formula cards, Malayalam support, module-wise questions, and a matching Master Answer Key. Only one selected section loads on screen.

Course Code: 3041 | **Category: Program Core** | **Credits: 3** | **L:T:P = 2:1:0** | **Theory Subject**

Lesson structure check

4
Official modules

45
Periods / semester

A4
Print-ready layout

1-click
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Course Overview

Subject title: Electric Circuits & Networks

Semester: 3 | **Course category:** Program Core | **Subject type:** Engineering Theory subject

This subject connects basic electrical laws, AC circuit behavior, network analysis, transformers, DC machines and AC machines. A student who studies this subject properly can read simple circuit diagrams, calculate circuit quantities, apply network theorems, and explain the working of common electrical machines used in electronics and industry.

Malayalam support: □ subject-□ electric circuit-□□□□ behavior, AC signal, RLC circuit, network theorem, transformer, DC machine, AC machine □□□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□ exam answer □□□□□□ □□□□□□□□□□□□□.

Source basis: Official SITTTR Revision 2021 syllabus content for course code 3041.

Course objective

To develop the skill to diagnose and rectify electric circuit networks and understand the operations of electrical machines.

Simple meaning: Circuit fault □□□□□□ □□□□□□□□□□□, circuit values □□□□□□ calculate □□□□□□□, motor/transformer working □□□□□□ explain □□□□□□□ — □□□□□ main aim.

Prerequisite knowledge

- Basic Engineering Mathematics principles from Mathematics I & II.
- Magnetism concepts from Applied Physics I & II.
- Ohm’s law, units, algebra, trigonometry and simple vector/phasor idea.

Course outcomes

CO	Outcome	Duration	Level
CO 1	Solve a given AC circuit to find various parameters using AC signals and component behavior.	9 hours	Applying

CO	Outcome	Duration	Level
CO 2	Apply network theorems for simplifying circuits and illustrate transformer operation.	14 hours	Applying
CO 3	Illustrate principles and operations of DC machines.	10 hours	Understanding
CO 4	Illustrate principles and operations of AC machines.	10 hours	Understanding

Module map

Module 1

AC signals, R-L-C behavior, phasor diagram, AC power, resonance, series and parallel RLC circuits.

Module 2

Ohm’s law, Kirchhoff’s laws, mesh and node analysis, network theorems, transformer principle and losses.

Module 3

DC generator, DC motor, back emf, starter, 3-point starter, DC motor comparison and characteristics.

Module 4

Alternator, AC motor classification, stepper motor, servo motor, universal motor, single-phase and three-phase induction motor.

How to study this subject

1. First learn definitions and units: RMS value, average value, frequency, reactance, impedance, power factor, resonance.
2. Draw every circuit and machine diagram in a separate notebook. Label current direction, voltage, winding, core and load.
3. For numerical problems, always write Given → Formula → Substitution → Calculation → Final answer with unit.
4. For theorem questions, write statement, circuit before simplification, circuit after simplification, calculation steps and result.
5. For machine questions, write principle, construction, working, equation, characteristics and applications.

Exam preparation method

Short answers

Definitions, symbols, units, laws, types, applications and advantages.

Problem solving

Practice AC power, impedance, resonance, Thevenin equivalent, maximum power transfer and transformer ratio.

Long answers

Use diagram + working + equation + application format. This gives better marks.

Malayalam tip: Diagram □□□□ question □□□□□□□□ diagram □□□□□ □□□□□□□□□□. □□□□□□ working principle □□□□□□. Formula □□□□□□ □□□□, explanation-□□□□ □□□□□ □□□□□□□□□□.

Quick navigation

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Mini Formula Bank

Use this as the first revision page before solving problems. Formula meaning and unit are important for marks.

Frequency

$$f = 1/T$$

f in Hz, T in seconds.

Angular frequency

$$\omega = 2\pi f$$

Used in reactance formulas.

RMS value

$$V_{rms} = V_m/\sqrt{2}$$

For sinusoidal waveform.

Average value

$$V_{avg} = 0.637V_m$$

Average of half cycle sine wave.

Inductive reactance

$$X_L = 2\pi fL$$

Increases with frequency.

Capacitive reactance

$$X_C = 1/(2\pi fC)$$

Decreases with frequency.

Series RLC impedance

$$Z = \sqrt{(R^2 + (X_L-X_C)^2)}$$

Unit: ohm.

Power factor

$$\cos\varphi = R/Z$$

Lagging for inductive circuit.

Active power

$$P = VI\cos\varphi$$

Unit: watt.

Reactive power

$$Q = VI\sin\varphi$$

Unit: VAR.

Apparent power

$$S = VI$$

Unit: VA.

Resonant frequency

$$f_r = 1/(2\pi\sqrt{LC})$$

At resonance $X_L = X_C$.

Thevenin resistance

$$R_{th}$$

Equivalent resistance seen from load terminals.

Maximum power

$$R_L = R_{th}$$

Condition for maximum power transfer.

Transformer ratio

$$V_1/V_2 = N_1/N_2$$

Ideal transformer relation.

Transformer EMF

$$E = 4.44\ f\ N\ \Phi_m$$

RMS induced emf.

DC generator EMF

$$E = P\Phi ZN\ /\ 60A$$

Generated emf equation.

Back EMF

$$E_b = V - I_a R_a$$

For DC motor.

Alternator EMF

$$E = 4.44\ k_w\ f\ \Phi\ T$$

Per phase emf.

Synchronous speed

$N_S = 120f/P$

rpm; P = number of poles.

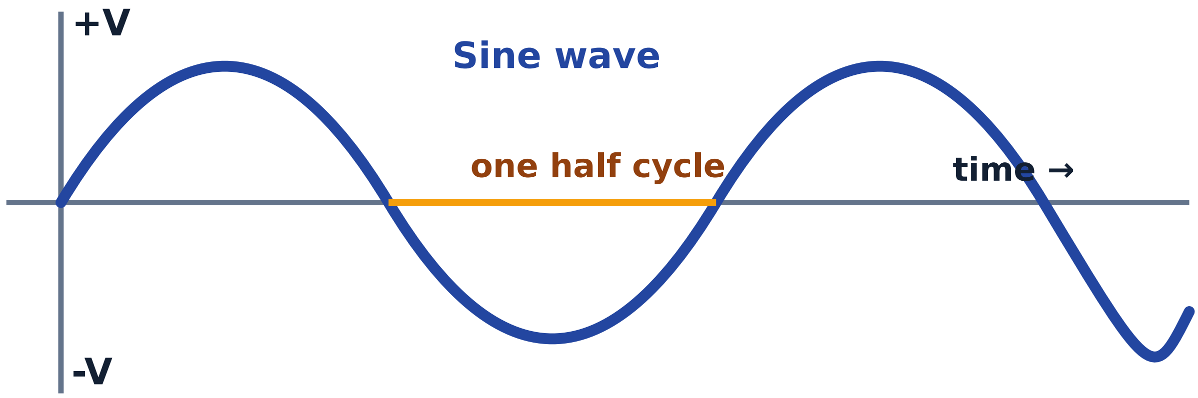
Module 1 — AC Signals and RLC Circuits

Learning outcome: Solve a given AC circuit to find various parameters using the concept of AC signals and the behavior of AC through different components.

Malayalam support: AC circuit problem □□□□□□□□□□□ waveform, RMS, average value, reactance, impedance, phase angle, power factor □□□□□□ clear □□□□□□□□□□□. R, L, C □□□□□□□□ current-□□□□ □□□□□□□□□□□□□ □□□□□□□□□□.

1.1 Alternating quantities

An alternating voltage or current changes its magnitude and direction periodically. A sine wave is the most common AC waveform because generation in alternators naturally gives sinusoidal voltage.



AC sine waveform: one complete cycle repeats after time period T.

Important definitions

- **Cycle:** One complete set of positive and negative values.
- **Time period:** Time taken for one cycle.

- **Frequency:** Number of cycles per second.
- **Amplitude:** Maximum value of the alternating quantity.
- **Phase:** Angular position of a waveform with respect to reference.
- **Form factor:** RMS value divided by average value.

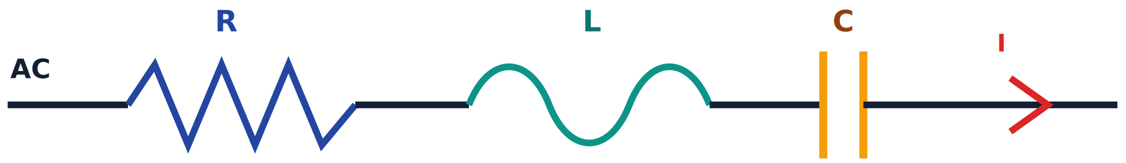
1.2 AC through R, L and C

Element	Current-voltage relation	Opposition	Power behavior
Pure Resistance	Voltage and current are in phase.	R	Consumes active power.
Pure Inductance	Current lags voltage by 90°.	$X_L = 2\pi fL$	Stores energy in magnetic field.
Pure Capacitance	Current leads voltage by 90°.	$X_C = 1/(2\pi fC)$	Stores energy in electric field.

Memory trick: Inductor-□ current □□□□□□□ □□□□: L = Lag. Capacitor-□ current □□□□□□□ □□□□: C = Current leads.

1.3 Series RLC circuit

In a series RLC circuit, the same current flows through R, L and C. The total opposition is called impedance. The inductive and capacitive reactances oppose each other, so the net reactance is $X_L - X_C$.



Series RLC circuit: same current flows through all components.

1.4 Solved example format

Example: Find impedance of a series RLC circuit

Given: $R = 30\ \Omega$, $X_L = 50\ \Omega$, $X_C = 10\ \Omega$.

Formula: $Z = \sqrt{R^2 + (X_L - X_C)^2}$

Substitution: $Z = \sqrt{30^2 + (50 - 10)^2} = \sqrt{900 + 1600}$

Calculation: $Z = \sqrt{2500} = 50\ \Omega$

Final answer: Impedance = $50\ \Omega$.

Expected Questions — Module 1

M1-Q1. Define cycle, time period, frequency, amplitude and phase of an AC waveform.

M1-Q2. Compare sine, square, triangular and sawtooth waveforms.

M1-Q3. Explain AC through pure resistance, pure inductance and pure capacitance.

M1-Q4. Draw and explain the phasor diagram of a series RLC circuit.

M1-Q5. Define active power, reactive power, apparent power and power factor.

M1-Q6. A series RLC circuit has $R = 40\ \Omega$, $X_L = 80\ \Omega$ and $X_C = 50\ \Omega$. Find impedance and power factor.

M1-Q7. Explain resonance and Q factor in an RLC circuit.

M1-Q8. List the steps to solve a parallel AC circuit problem.

Module 1 checklist

AC terms · RMS · Reactance · Impedance · Power factor · Resonance

Common mistake: Do not add X_L and X_C directly in series RLC. Use net reactance $X_L - X_C$.

Module 2 — Network Theorems and Transformers

Learning outcome: Apply network theorems for simplifying electric circuits and illustrate transformer operation.

Malayalam support: □□□□ circuit □□□□□ equivalent circuit □□□□□ solve □□□□□□□□□□□□□ network theorem-□□□□□ main use. Transformer AC voltage step-up/step-down □□□□□□□ □□□□□□□□□□□□□□.

2.1 Laws and analysis methods

Method	Use	Exam writing point
Ohm’s law	Relates voltage, current and resistance.	V = IR
Kirchhoff’s Current Law	Current relation at a node.	Sum of incoming currents = sum of outgoing currents.
Kirchhoff’s Voltage Law	Voltage relation in a closed loop.	Algebraic sum of voltages in a loop is zero.
Mesh analysis	Solves circuits by loop currents.	Best for planar circuits with loops.
Node analysis	Solves circuits by node voltages.	Best when many current sources are present.

2.2 Network theorems

Superposition theorem

In a linear bilateral network, current or voltage in an element is the algebraic sum of effects due to each independent source acting alone.

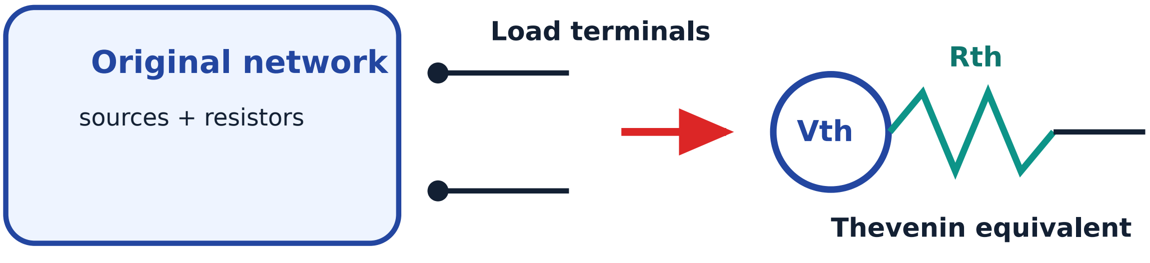
Thevenin’s theorem

Any linear two-terminal network can be replaced by a voltage source V_{th} in series with resistance R_{th}.

Maximum power transfer

Maximum power is transferred to the load when load resistance equals Thevenin resistance.

2.3 Thevenin equivalent visual



Thevenin theorem replaces a complex network with Vth and Rth.

2.4 Transformer basics

A transformer is a static AC device that transfers electrical energy from one circuit to another by mutual induction. It can increase or decrease AC voltage without changing frequency.

Construction

- Laminated magnetic core
- Primary winding
- Secondary winding
- Insulation and tank/cooling arrangement

Losses

- Copper loss in windings
- Hysteresis loss in core
- Eddy current loss in core
- Leakage flux related losses

Malayalam support: Transformer DC-□ work □□□□□□□□□□. □□□□□ mutual induction □□□□□□□□□ changing flux □□□□. DC steady □□□□ changing flux □□□□□□□□□□□.

Expected Questions — Module 2

M2-Q1. State Ohm’s law, Kirchhoff’s Current Law and Kirchhoff’s Voltage Law.

M2-Q2. Compare mesh analysis and node analysis.

M2-Q3. State and explain Superposition theorem.

M2-Q4. State Thevenin’s theorem and write the steps to find Thevenin equivalent circuit.

M2-Q5. State Maximum Power Transfer theorem and its condition.

M2-Q6. Explain the working principle and construction of a transformer.

M2-Q7. Derive the emf equation of a transformer.

M2-Q8. List transformer losses, types and applications.

Module 2 checklist

Ohm **KCL** **KVL** **Mesh** **Node** **Thevenin** **Transformer**

For theorem answers: statement → circuit → steps → result → application.

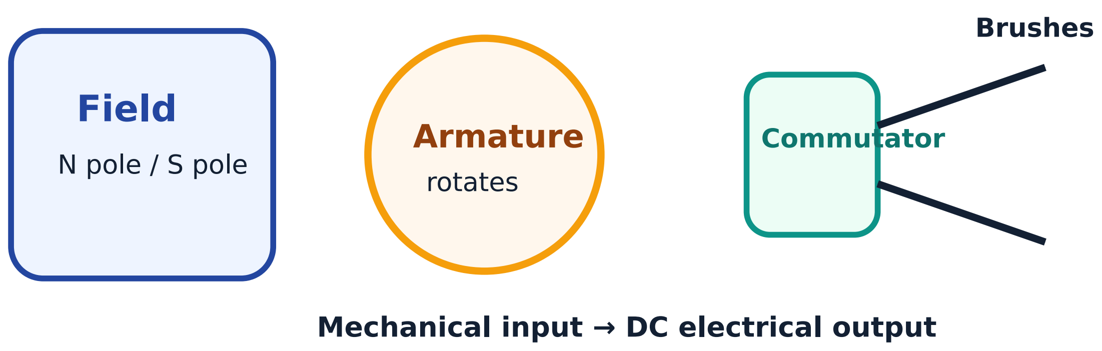
Module 3 — DC Machines

Learning outcome: Illustrate the principles and operations of DC generators and DC motors.

Malayalam support: DC generator mechanical energy-□□ DC electrical energy □□□□□□□□□□. DC motor DC electrical energy-□□ mechanical energy □□□□□□□□□□. □□□□□□□□□ magnetic field + conductor + motion main idea □□□.

3.1 DC generator

A DC generator works on Faraday’s law of electromagnetic induction. When an armature conductor cuts magnetic flux, emf is induced in the conductor. The commutator converts the internally generated alternating emf into unidirectional output at the brushes.



Basic blocks of a DC generator.

3.2 DC motor

A DC motor works on the principle that a current-carrying conductor placed in a magnetic field experiences a mechanical force. Back emf is generated in the armature and opposes the applied voltage. At starting, back emf is zero, so starter resistance is required to limit armature current.

Motor type	Characteristic	Common application
DC shunt motor	Nearly constant speed	Lathes, fans, blowers
DC series motor	High starting torque	Traction, cranes, hoists
DC compound motor	Good starting torque with better speed regulation	Presses, elevators, heavy machines

3.3 Starter in DC motor

A starter is used to limit the high starting current of a DC motor. It also provides no-voltage protection and overload protection in practical systems. A 3-point starter has line, armature and field terminals.

Common exam mistake: Do not write that starter increases speed. The main purpose is to limit starting current and protect the motor.

Expected Questions — Module 3

M3-Q1. Explain the working principle of a DC generator.

M3-Q2. Write the emf equation of a DC generator and explain each term.

M3-Q3. Explain armature reaction in a DC generator.

M3-Q4. Explain the working principle of a DC motor.

M3-Q5. What is back emf? Explain its significance in a DC motor.

M3-Q6. Why is a starter necessary in a DC motor?

M3-Q7. Explain the working of a 3-point starter.

M3-Q8. Compare DC shunt, series and compound motors with characteristics and applications.

Module 3 checklist

Generator **EMF equation** **Armature reaction** **Motor** **Back EMF** **Starter**

Module 4 — AC Machines

Learning outcome: Illustrate the principles and operations of alternators and AC motors.

Malayalam support: Alternator AC generate □□□□□□□□□□□. AC motors AC supply □□□□□□□□□□□ rotating magnetic field □□□□□□□□□□ mechanical output □□□□□□□□□.

4.1 Alternator

An alternator converts mechanical energy into AC electrical energy. The rotor produces magnetic field and the stator contains armature windings. When the rotor rotates, the magnetic flux linking the stator conductors changes and emf is induced.

Important equations

- Frequency relation: $f = PN / 120$
- Synchronous speed: $N_s = 120f / P$
- EMF equation: $E = 4.44\ k_w\ f\ \Phi\ T$

Open circuit characteristic

OCC shows the relation between generated emf and field current at constant speed. It is also called magnetization characteristic.

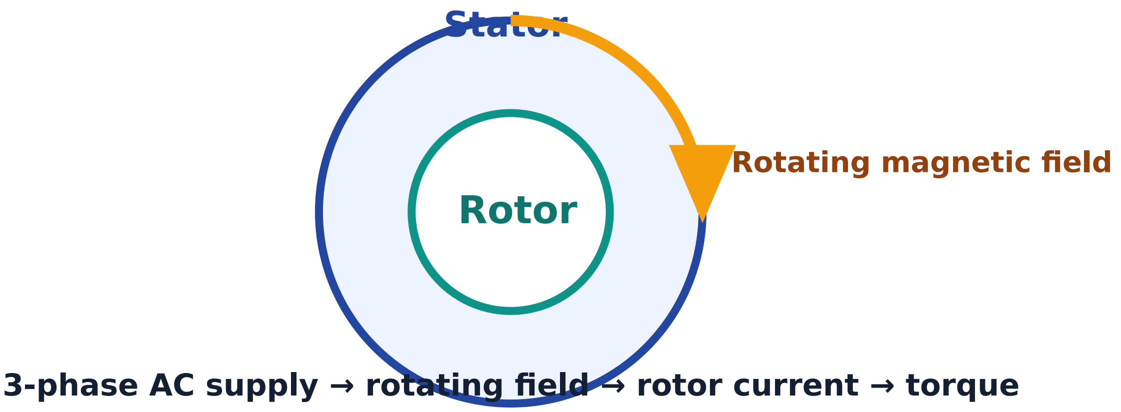
4.2 AC motors

Motor	Working idea	Application
Stepper motor	Moves in fixed angular steps.	Printers, CNC positioning, robotics.
Servo motor	Feedback-controlled motor for precise position or speed.	Automation, antenna control, robotics.
Universal motor	Can work on AC or DC; high speed.	Drills, mixers, vacuum cleaners.
Single-phase induction motor	Uses single-phase supply; needs starting arrangement.	Fans, pumps, small machines.

Motor	Working idea	Application
Three-phase induction motor	Uses rotating magnetic field produced by three-phase supply.	Industrial drives, compressors, conveyors.

4.3 Three-phase induction motor principle

When a balanced three-phase supply is given to the stator winding, a rotating magnetic field is produced. This rotating field cuts the rotor conductors and induces emf. Since the rotor circuit is closed, current flows in the rotor. Interaction between the rotating field and rotor current produces torque.



Working principle of a three-phase induction motor.

Expected Questions — Module 4

M4-Q1. Explain the working principle of an alternator.

M4-Q2. Write the emf equation of an alternator and explain the terms.

M4-Q3. Define synchronous speed and derive the relation $N_s = 120f/P$.

M4-Q4. What is open circuit characteristic of an alternator?

M4-Q5. Classify AC motors.

M4-Q6. Explain the working and applications of stepper motor, servo motor and universal motor.

M4-Q7. Explain the working principle of a single-phase induction motor.

M4-Q8. Explain the working principle and applications of a three-phase induction motor.

Module 4 checklist

Alternator **EMF** **Ns** **OCC** **Stepper** **Servo** **Induction motor**

One-page Quick Revision

Module 1 must-know

- AC terms and waveforms
- RMS and average values
- R, L and C behavior
- Impedance, power factor and resonance

Module 2 must-know

- Ohm’s law, KCL, KVL
- Mesh and node analysis
- Superposition, Thevenin, maximum power transfer
- Transformer principle, emf equation, losses and applications

Module 3 must-know

- DC generator principle and emf equation
- Armature reaction
- DC motor principle and back emf
- Need of starter and 3-point starter

Module 4 must-know

- Alternator principle and emf equation
- Synchronous speed
- AC motor classification
- Single-phase and three-phase induction motor working

Important definitions

Term	Quick meaning
Impedance	Total opposition offered by an AC circuit.
Power factor	Cosine of phase angle between voltage and current.
Resonance	Condition in RLC circuit when $X_L = X_C$.
Thevenin equivalent	Voltage source V_{th} in series with resistance R_{th}.
Back emf	Emf induced in DC motor armature opposing applied voltage.
Synchronous speed	Speed of rotating magnetic field in an AC machine.

Common exam mistakes

- Writing only formula without explaining each symbol.
- Forgetting units in numerical answers.

- Confusing inductive and capacitive phase relation.
- Writing theorem statement but not writing solving steps.
- Drawing machine diagrams without labels.
- Writing transformer works on DC supply. Transformer requires AC/changing flux.

Last-minute checklist

1. Revise formula bank and write each formula once.
2. Practice one impedance problem and one Thevenin problem.
3. Redraw transformer, DC generator and three-phase induction motor diagrams.
4. Revise applications of all motors.
5. Read the Master Answer Key for exact answer point format.

Master Answer Key

Answers below match the Expected Questions shown in Module 1 to Module 4. The question numbering is kept exactly same.

Module 1 Answers

M1-Q1.

Cycle: one complete positive and negative variation. **Time period:** time for one cycle. **Frequency:** cycles per second. **Amplitude:** maximum value. **Phase:** angular position of waveform with respect to reference.

M1-Q2.

Sine wave changes smoothly and is common in AC generation. Square wave switches between two levels. Triangular wave rises and falls linearly. Sawtooth wave rises gradually and falls suddenly or vice versa. Mention shape, harmonic content and common use.

M1-Q3.

In pure resistance, V and I are in phase and power is consumed. In pure inductance, current lags voltage by 90° and energy is stored in magnetic field. In pure capacitance, current leads voltage by 90° and energy is stored in electric field.

M1-Q4.

Draw current as reference. Draw V_R in phase with current, V_L leading current by 90°, and V_C lagging current by 90°. Resultant supply voltage is vector sum of V_R and net reactive voltage.

M1-Q5.

Active power P = VIcosφ in watts. Reactive power Q = VIsinφ in VAR. Apparent power S = VI in VA. Power factor is cosφ, the cosine of the angle between voltage and current.

M1-Q6.

Given: R = 40 Ω, X_L = 80 Ω, X_C = 50 Ω. Net reactance = 30 Ω. Z = $\sqrt{(40^2 + 30^2)}$ = $\sqrt{2500}$ = 50 Ω. Power factor = R/Z = 40/50 = 0.8. Since $X_L > X_C$, power factor is lagging.

M1-Q7.

Resonance is the condition in an RLC circuit when $X_L = X_C$. At resonance, impedance is minimum in a series circuit and current is maximum. Q factor indicates sharpness of resonance and voltage/current magnification.

M1-Q8.

Steps: identify branches, find branch impedance/admittance, calculate branch currents, resolve currents into active and reactive components, add components, find total current, power factor and power.

Module 2 Answers

M2-Q1.

Ohm's law: current through a conductor is directly proportional to voltage when physical conditions are constant, V = IR. KCL: algebraic sum of currents at a node is zero. KVL: algebraic sum of voltages around a closed loop is zero.

M2-Q2.

Mesh analysis uses loop currents and KVL. It is suitable for planar circuits. Node analysis uses node voltages and KCL. It is useful when current sources are present and the circuit has fewer essential nodes.

M2-Q3.

Superposition theorem states that in a linear bilateral network with multiple independent sources, current or voltage in an element is the algebraic sum of effects produced by each source acting alone. While considering one source, replace other voltage sources by short circuit and current sources by open circuit.

M2-Q4.

Thevenin's theorem: any linear two-terminal network can be replaced by an equivalent voltage source V_{th} in series with resistance R_{th}. Steps: remove load, find open-circuit voltage V_{th}, deactivate independent sources and find R_{th}, reconnect load to equivalent circuit and solve.

M2-Q5.

Maximum Power Transfer theorem states that maximum power is delivered to load when load resistance equals internal/Thevenin resistance of source network. Condition: $R_L = R_{th}$.

M2-Q6.

A transformer works on mutual induction. AC in primary creates alternating flux in core. This flux links secondary winding and induces emf. Construction includes laminated core, primary winding, secondary winding, insulation and cooling arrangement.

M2-Q7.

For sinusoidal flux, RMS induced emf per winding is $E = 4.44\ f\ N\ \Phi_m$. For primary, $E_1 = 4.44\ f\ N_1\ \Phi_m$. For secondary, $E_2 = 4.44\ f\ N_2\ \Phi_m$. Explain f = frequency, N = turns, Φ_m = maximum flux.

M2-Q8.

Losses: copper loss, hysteresis loss, eddy current loss and leakage losses. Types: step-up, step-down, isolation, core type, shell type, power and distribution transformers. Applications: power transmission, adapters, isolation, measurement and electronic power supplies.

Module 3 Answers

M3-Q1.

A DC generator works on Faraday’s law. When armature conductors rotate in a magnetic field, they cut magnetic flux and emf is induced. The commutator converts internally alternating emf into unidirectional DC at the brushes.

M3-Q2.

Generated emf $E = P\Phi ZN / 60A$, where P = number of poles, Φ = flux per pole, Z = total armature conductors, N = speed in rpm, A = number of parallel paths.

M3-Q3.

Armature reaction is the effect of armature flux on the main field flux. It distorts and weakens main flux, shifts neutral plane and can cause sparking at brushes. It is reduced using interpoles or compensating windings.

M3-Q4.

A DC motor works on the principle that a current-carrying conductor placed in a magnetic field experiences force. The direction of force is given by Fleming’s left-hand rule. This force produces torque and rotates the armature.

M3-Q5.

Back emf is the emf induced in the rotating armature of a DC motor that opposes the applied voltage. Significance: it limits armature current during running and makes the motor self-regulating with load.

M3-Q6.

At starting, back emf is zero, so armature current can become very high because armature resistance is small. A starter adds external resistance at starting to limit current and provides protection.

M3-Q7.

A 3-point starter has line, armature and field terminals. Starting resistance is gradually cut out as motor speed and back emf increase. It includes no-voltage release and overload release for protection.

M3-Q8.

DC shunt motor: nearly constant speed, used in fans and lathes. DC series motor: high starting torque, used in traction and cranes. DC compound motor: combines good torque and speed regulation, used in elevators and heavy machines.

Module 4 Answers

M4-Q1.

An alternator converts mechanical energy into AC electrical energy. The rotor field rotates and cuts stator conductors, inducing alternating emf in stator windings according to electromagnetic induction.

M4-Q2.

Alternator emf per phase $E = 4.44\ k_w\ f\ \Phi\ T$, where k_w = winding factor, f = frequency, Φ = flux per pole and T = series turns per phase.

M4-Q3.

Synchronous speed is the speed of rotating magnetic field. In one mechanical revolution, P/2 cycles are produced. Hence $f = PN/120$ and $N_s = 120f/P$ rpm.

M4-Q4.

Open circuit characteristic of an alternator is the graph between generated emf and field current at constant speed with no load connected. It shows magnetization behavior and saturation.

M4-Q5.

AC motors can be classified as synchronous motors and induction motors. Induction motors may be single-phase or three-phase. Special AC motors include stepper motor, servo motor and universal motor.

M4-Q6.

Stepper motor moves in fixed angular steps and is used in positioning. Servo motor uses feedback for precise control and is used in automation. Universal motor works on AC or DC, has high speed and is used in portable tools and appliances.

M4-Q7.

A single-phase induction motor works by induction but is not self-starting because single-phase supply produces pulsating magnetic field. Starting methods such as split-phase or capacitor start create phase difference to start rotation.

M4-Q8.

A three-phase induction motor works on rotating magnetic field. Three-phase stator supply creates rotating field, which induces current in rotor conductors. Interaction of rotor current and rotating field produces torque. Applications include pumps, compressors, conveyors and industrial drives.